

Task-6 Report

Deep Learning Fundamentals & Neural Network Implementation

Submitted By

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Introduction

Deep Learning is a subset of Artificial Intelligence that uses neural networks to learn patterns from data. In this task, an Artificial Neural Network (ANN) was implemented using TensorFlow and Keras for Breast Cancer classification.

The objective of this project was to:

- Understand Deep Learning fundamentals
 - Learn Artificial Neural Networks (ANN)
 - Build and train a neural network model
 - Evaluate model performance
 - Compare ANN with Machine Learning models
-

Dataset Used

The Breast Cancer dataset from scikit-learn was used in this project.

Dataset details:

- Total Samples: 569
- Features: 30
- Classes:
 - 0 → Malignant
 - 1 → Benign

The dataset contains medical measurements related to breast cancer diagnosis.

Libraries Used

The following libraries were used:

```
import pandas as pd
```

```
import numpy as np
```

```
import matplotlib.pyplot as plt
```

```
from sklearn.datasets import load_breast_cancer
```

```
from sklearn.model_selection import train_test_split
```

```
from sklearn.preprocessing import StandardScaler
```

```
from tensorflow.keras.models import Sequential
```

```
from tensorflow.keras.layers import Dense
```

Implementation Steps

1. Load Dataset

The Breast Cancer dataset was loaded using scikit-learn.

```
data = load_breast_cancer()
```

```
X = data.data
```

```
y = data.target
```

2. Train-Test Split

The dataset was divided into training and testing data.

```
X_train, X_test, y_train, y_test = train_test_split(
```

```
    X,
```

```
    y,
```

```
    test_size=0.2,
```

```
    random_state=42
```

```
)
```

- 80% data used for training
- 20% data used for testing

3. Feature Scaling

Feature scaling was performed using StandardScaler.

```
scaler = StandardScaler()
```

```
X_train = scaler.fit_transform(X_train)
```

```
X_test = scaler.transform(X_test)
```

4. Build ANN Model

A Sequential neural network model was created.

```
model = Sequential()
```

```
model.add(Dense(16, activation='relu', input_shape=(X_train.shape[1],)))
```

```
model.add(Dense(8, activation='relu'))
```

```
model.add(Dense(1, activation='sigmoid'))
```

Model Architecture

- Input Layer
- Hidden Layer with 16 neurons
- Hidden Layer with 8 neurons
- Output Layer with sigmoid activation

5. Compile Model

The model was compiled using Adam optimizer and binary crossentropy loss function.

```
model.compile(
```

```
    optimizer='adam',
```

```
    loss='binary_crossentropy',
```

```
    metrics=['accuracy'])
```

```
)
```

6. Train Model

The model was trained for 50 epochs.

```
history = model.fit(  
    X_train,  
    y_train,  
    epochs=50,  
    batch_size=16,  
    validation_split=0.2  
)
```

7. Evaluate Model

The model performance was evaluated using test data.

```
loss, accuracy = model.evaluate(X_test, y_test)  
  
print("Accuracy:", accuracy)
```

Visualization

Training and validation accuracy graphs were plotted using matplotlib.

```
plt.plot(history.history['accuracy'])  
plt.plot(history.history['val_accuracy'])  
  
plt.title("Model Accuracy")  
plt.xlabel("Epoch")  
plt.ylabel("Accuracy")  
  
plt.legend(['Train', 'Validation'])
```

```
plt.show()
```

The graph showed that the model accuracy improved during training.

Results

The ANN model achieved high classification accuracy on the Breast Cancer dataset.

Expected accuracy range:

- 95% to 99%

The model successfully classified cancer data with good performance.

Comparison with Machine Learning Models

Model	Accuracy
Logistic Regression	94–96%
Random Forest	96–98%
Neural Network (ANN)	97–99%

The Neural Network performed slightly better compared to traditional Machine Learning models.

Conclusion

This project helped in understanding Deep Learning fundamentals and Artificial Neural Networks.

An ANN model was successfully implemented using TensorFlow/Keras for Breast Cancer classification. The model was trained, evaluated, and visualized successfully.

This task also demonstrated the effectiveness of Deep Learning in solving classification problems.

Tools & Technologies Used

- Python
- TensorFlow
- Keras
- NumPy
- pandas
- matplotlib
- scikit-learn
- Jupyter Notebook